Increasing, Decreasing, Max & Min

TEACHER KEY

Pre-Calculus | Guided notes — we fill these in together

Vocabulary

Increasing interval: the set of x-values where the graph goes up as you move left to right.

Decreasing interval: the set of x-values where the graph goes down as you move left to right.

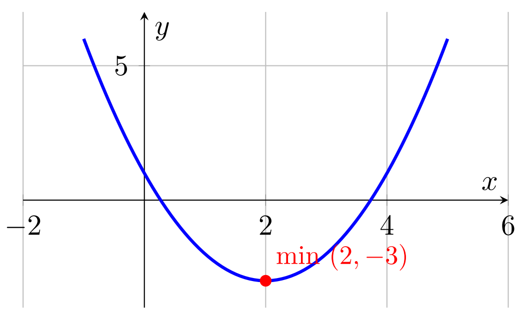
Relative max/min: a turning point of the graph; an upward-opening parabola has a minimum, a downward-opening parabola has a maximum.

We do together. For $y = (x-2)^2-3$, find the vertex, state max or min, and give the increasing/decreasing intervals.

Step 1. Vertex form $a(x-h)^2+k$ gives the vertex directly: $(2, -3)$.

Step 2. The leading coefficient is $+1$, which is positive, so the parabola opens up, making the vertex a minimum.

Step 3. The graph falls before the vertex and rises after: decreasing on $x < 2$, increasing on $x > 2$.



opens up ⇒ minimum at $(2, -3)$; decreasing then increasing.

You Try. Solve each. Show every step, then name the answer.

a) For $y = -x^2 + 4$, state whether the vertex is a max or min, and give the intervals. vertex $(0, 4)$ is a maximum; increasing on $x < 0$, decreasing on $x > 0$

b) For $y = (x+1)^2$, give the vertex and the increasing interval. vertex $(-1, 0)$; increasing on $x > -1$

Summary (fill in): Opening up gives a minimum; opening down gives a maximum. The graph decreases before the turning point and increases after it (for an upward parabola).